Important Applications of Arrays

a) storing and manipulating of vectors / matrices / data

b) calculating statistics (mean, standard deviation, median, etc.)

c) strings / text (arrays of characters)

d) processing experimental data, plotting, graphics, audio, etc.

e) discrete approximation of functions (computer solution of calculus problems, etc.)

Vectors and Matrices

review the material in matrix_algebra_review if needed so you know basic matrix algebra (+, -, *, transpose)

vectors and matrices are not variable types in C++, however 1D and 2D arrays can be used to represent them

most vectors / matrices have real components so arrays of doubles or floats are normally used (unless integer is specified)

one problem is that vectors / matrices normally start at index 1, whereas C++ arrays start at index 0, eg

double A[3] -- goes from A[0] to A[2] -- index 0 array

3x1 vector A -- goes from A[1] to A[3] -- index 1 array

a problem since most math formulas, etc. start at index 1

a simple approach to address this problem is to declare an array A with one extra element and not use index 0, eg

double A[3+1] -- goes from A[0] to A[3]

use this array from A[1] to A[3] to store vector A (ignore A[0])

analogous to having an extra room in your house you never use

we can use the same approach for matrices / 2D arrays, eg

2x3 matrix B -- goes from B[1][1] to B[2][3]

double B[2+1][3+1] -- goes from B[0][0] to B[2][3]

use this array from B[1][1] to B[2][3] to store matrix B
(ignore row 0, B[0][j] and col 0, B[i][0])

matrix algebra (+, *, etc.) is not built into C++ so you need to perform the operations on individual elements using for loops